Restore & Spin
12-mode restoration manual

Select a mode on the Power App/Page, then use the matching procedure to restore the physical Pi panel.

Success signal: LED4 turns green.

Find your procedure

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Printing: US Letter, portrait, actual size (100%). For A4 paper, choose Fit. Black-and-white printing is supported. Each mode occupies one page.

Getting started

Know the controls

Panel label	Control	How to use it
SW1 / SW2	Safety switches 1 / 2	Set ON (closed) or OFF (open).
SW3 / SW4	Toggle switches 1 / 2	Set ON (closed) or OFF (open).
SW5 / SW6	Keyed switches 1 / 2	Set ON (closed) or OFF (open).
J1 / J2 / J3 / J4	Panel sockets	Plug in (connected) or unplug (open).

Read the indicator lights

Indicator	Meaning when lit
LED1 - Blue	Ready/power condition is active.
LED2 - Yellow	Restoration is still in progress.
LED3 - Orange	At least one socket is connected.
LED4 - Green	**The selected mode is restored.**

Before each run

1. Select the intended mode in the Power App/Page.

2. To reset the fault for a new player, put each control into the opposite of its target state on the selected mode page.

3. Ask the player to follow steps 1-10 in order.

4. Confirm LED4 is green before starting the next mode.

If LED4 is not green, confirm the selected mode and check all ten controls against that mode's procedure.

Smart Thermostat

Select **Smart Thermostat** in the Power App/Page. Follow these steps in order.

Step	Control and function	Action / target
1	**SW3** Toggle switch 1 - fan running	Set **ON** Closed
2	**J1** Socket J1 connected	Plug in Connected
3	**SW1** Safety switch 1 - power	Set **ON** Closed
4	**J4** Socket J4 connected	Unplug Open
5	**SW5** Keyed switch 1	Set **ON** Closed
6	**J2** Socket J2 connected	Plug in Connected
7	**SW2** Safety switch 2 - eco lockout	Set **OFF** Open
8	**SW6** Keyed switch 2	Set **OFF** Open
9	**J3** Socket J3 connected	Unplug Open
10	**SW4** Toggle switch 2 - heat/cool select	Set **OFF** Open

Completion check: LED4 is green.

Thermostat restored - holding your set temperature.

Mode reference: thermostat

Smart Lock & Security Panel

Select **Smart Lock & Security Panel** in the Power App/Page. Follow these steps in order.

Step	Control and function	Action / target
1	**J2** Socket J2 connected	Unplug Open
2	**SW5** Keyed switch 1 - service key	Set **ON** Closed
3	**SW2** Safety switch 2 - deadbolt	Set **ON** Closed
4	**J4** Socket J4 connected	Plug in Connected
5	**SW1** Safety switch 1 - power	Set **ON** Closed
6	**SW4** Toggle switch 2 - alarm arm	Set **ON** Closed
7	**J1** Socket J1 connected	Unplug Open
8	**SW3** Toggle switch 1 - auto-lock	Set **OFF** Open
9	**SW6** Keyed switch 2	Set **ON** Closed
10	**J3** Socket J3 connected	Plug in Connected

Completion check: LED4 is green.

House is secured now.

Mode reference: smart-lock

Cold-Chain Warehouse Sensor

Select **Cold-Chain Warehouse Sensor** in the Power App/Page. Follow these steps in order.

Step	Control and function	Action / target
1	**SW4** Toggle switch 2 - cloud sync	Set **ON** Closed
2	**J3** Socket J3 connected	Plug in Connected
3	**SW1** Safety switch 1 - power	Set **ON** Closed
4	**SW6** Keyed switch 2	Set **OFF** Open
5	**J1** Socket J1 connected	Plug in Connected
6	**SW2** Safety switch 2 - calibration	Set **ON** Closed
7	**J4** Socket J4 connected	Unplug Open
8	**SW5** Keyed switch 1	Set **OFF** Open
9	**SW3** Toggle switch 1 - threshold	Set **OFF** Open
10	**J2** Socket J2 connected	Plug in Connected

Completion check: LED4 is green.

Cold storage is stable and monitoring is back online.

Mode reference: coldchain-sensor

Water Pump Station Controller

Select **Water Pump Station Controller** in the Power App/Page. Follow these steps in order.

Step	Control and function	Action / target
1	**J4** Socket J4 connected	Plug in Connected
2	**SW2** Safety switch 2 - pump enable	Set **ON** Closed
3	**SW5** Keyed switch 1	Set **OFF** Open
4	**J1** Socket J1 connected	Unplug Open
5	**SW4** Toggle switch 2 - remote override	Set **OFF** Open
6	**J2** Socket J2 connected	Plug in Connected
7	**SW1** Safety switch 1 - power	Set **ON** Closed
8	**J3** Socket J3 connected	Unplug Open
9	**SW6** Keyed switch 2	Set **ON** Closed
10	**SW3** Toggle switch 1 - pressure relief	Set **ON** Closed

Completion check: LED4 is green.

Pump station is back online and operating normally.

Mode reference: water-pump

Tactical Field Radio

Select **Tactical Field Radio** in the Power App/Page. Follow these steps in order.

Step	Control and function	Action / target
1	**SW6** Keyed switch 2 - crypto key	Set **ON** Closed
2	**J2** Socket J2 connected	Plug in Connected
3	**SW3** Toggle switch 1 - channel hop	Set **ON** Closed
4	**J1** Socket J1 connected	Unplug Open
5	**SW1** Safety switch 1 - power	Set **ON** Closed
6	**J4** Socket J4 connected	Unplug Open
7	**SW2** Safety switch 2 - encryption	Set **ON** Closed
8	**SW5** Keyed switch 1	Set **OFF** Open
9	**J3** Socket J3 connected	Plug in Connected
10	**SW4** Toggle switch 2 - transmit	Set **ON** Closed

Completion check: LED4 is green.

Secure link established with base.

Mode reference: field-radio

Smart EV Charger

Select **Smart EV Charger** in the Power App/Page. Follow these steps in order.

Step	Control and function	Action / target
1	**J3** Socket J3 connected	Unplug Open
2	**SW1** Safety switch 1 - power	Set **ON** Closed
3	**SW5** Keyed switch 1 - service lock	Set **ON** Closed
4	**SW4** Toggle switch 2 - billing link	Set **ON** Closed
5	**J1** Socket J1 connected	Plug in Connected
6	**SW2** Safety switch 2 - contactor	Set **ON** Closed
7	**J4** Socket J4 connected	Plug in Connected
8	**SW6** Keyed switch 2	Set **OFF** Open
9	**SW3** Toggle switch 1 - ground check	Set **ON** Closed
10	**J2** Socket J2 connected	Unplug Open

Completion check: LED4 is green.

EV charger restored - charging resumed.

Mode reference: ev-charger

Elevator Control Panel

Select **Elevator Control Panel** in the Power App/Page. Follow these steps in order.

Step	Control and function	Action / target
1	**SW2** Safety switch 2 - brake release	Set **ON** Closed
2	**J4** Socket J4 connected	Unplug Open
3	**SW6** Keyed switch 2 - inspection key	Set **ON** Closed
4	**J2** Socket J2 connected	Plug in Connected
5	**SW1** Safety switch 1 - power	Set **ON** Closed
6	**SW3** Toggle switch 1 - doors closed	Set **ON** Closed
7	**J1** Socket J1 connected	Plug in Connected
8	**SW5** Keyed switch 1	Set **OFF** Open
9	**J3** Socket J3 connected	Unplug Open
10	**SW4** Toggle switch 2 - inspection mode	Set **OFF** Open

Completion check: LED4 is green.

Elevator back in service.

Mode reference: elevator

Traffic Signal Controller

Select **Traffic Signal Controller** in the Power App/Page. Follow these steps in order.

Step	Control and function	Action / target
1	**J1** Socket J1 connected	Plug in Connected
2	**SW4** Toggle switch 2 - manual override	Set **OFF** Open
3	**SW1** Safety switch 1 - power	Set **ON** Closed
4	**J3** Socket J3 connected	Plug in Connected
5	**SW5** Keyed switch 1 - cabinet key	Set **ON** Closed
6	**SW2** Safety switch 2 - controller	Set **ON** Closed
7	**J2** Socket J2 connected	Unplug Open
8	**SW6** Keyed switch 2	Set **OFF** Open
9	**SW3** Toggle switch 1 - timing plan	Set **ON** Closed
10	**J4** Socket J4 connected	Plug in Connected

Completion check: LED4 is green.

Traffic signal restored to normal cycle.

Mode reference: traffic-signal

Hospital Patient Monitor

Select **Hospital Patient Monitor** in the Power App/Page. Follow these steps in order.

Step	Control and function	Action / target
1	**SW5** Keyed switch 1 - nurse key	Set **ON** Closed
2	**J3** Socket J3 connected	Unplug Open
3	**SW2** Safety switch 2 - leads connected	Set **ON** Closed
4	**J1** Socket J1 connected	Plug in Connected
5	**SW4** Toggle switch 2 - alarm audible	Set **ON** Closed
6	**SW1** Safety switch 1 - power	Set **ON** Closed
7	**J4** Socket J4 connected	Unplug Open
8	**SW3** Toggle switch 1 - SpO2 sensor	Set **ON** Closed
9	**SW6** Keyed switch 2	Set **OFF** Open
10	**J2** Socket J2 connected	Plug in Connected

Completion check: LED4 is green.

Patient monitor restored - vitals streaming normally.

Mode reference: patient-monitor

Retail Payment Kiosk

Select **Retail Payment Kiosk** in the Power App/Page. Follow these steps in order.

Step	Control and function	Action / target
1	**SW3** Toggle switch 1 - card reader	Set **ON** Closed
2	**SW6** Keyed switch 2 - cash drawer key	Set **ON** Closed
3	**J4** Socket J4 connected	Plug in Connected
4	**SW1** Safety switch 1 - power	Set **ON** Closed
5	**J2** Socket J2 connected	Unplug Open
6	**SW4** Toggle switch 2 - network	Set **ON** Closed
7	**J1** Socket J1 connected	Unplug Open
8	**SW2** Safety switch 2 - receipt printer	Set **ON** Closed
9	**J3** Socket J3 connected	Plug in Connected
10	**SW5** Keyed switch 1	Set **OFF** Open

Completion check: LED4 is green.

Payment kiosk back online and processing.

Mode reference: retail-kiosk

Drone Ground Control

Select **Drone Ground Control** in the Power App/Page. Follow these steps in order.

Step	Control and function	Action / target
1	**J2** Socket J2 connected	Plug in Connected
2	**SW4** Toggle switch 2 - uplink	Set **ON** Closed
3	**SW6** Keyed switch 2 - flight key	Set **ON** Closed
4	**J1** Socket J1 connected	Plug in Connected
5	**SW2** Safety switch 2 - GPS lock	Set **ON** Closed
6	**J3** Socket J3 connected	Unplug Open
7	**SW1** Safety switch 1 - power	Set **ON** Closed
8	**SW5** Keyed switch 1	Set **OFF** Open
9	**J4** Socket J4 connected	Plug in Connected
10	**SW3** Toggle switch 1 - telemetry	Set **ON** Closed

Completion check: LED4 is green.

Drone link restored - ready for flight ops.

Mode reference: drone-ground-control

Industrial Robot Arm

Select **Industrial Robot Arm** in the Power App/Page. Follow these steps in order.

Step	Control and function	Action / target
1	**J4** Socket J4 connected	Unplug Open
2	**SW1** Safety switch 1 - main power	Set **ON** Closed
3	**SW6** Keyed switch 2 - teach pendant key	Set **ON** Closed
4	**J2** Socket J2 connected	Plug in Connected
5	**SW3** Toggle switch 1 - servo enable	Set **ON** Closed
6	**J1** Socket J1 connected	Unplug Open
7	**SW5** Keyed switch 1 - cell access	Set **ON** Closed
8	**SW2** Safety switch 2 - E-stop reset	Set **ON** Closed
9	**J3** Socket J3 connected	Plug in Connected
10	**SW4** Toggle switch 2 - guard door	Set **OFF** Open

Completion check: LED4 is green.

Robot cell restored - supervised automation is ready.

Mode reference: industrial-robot-arm

Demo cable build reference

For the person preparing the demo cables. Cable wiring is independent of the Pi harness and does not change the panel receptacle wiring.

Side A and Side B identify the two ends of each cable. The contact pairs below reproduce the manual's cable reference.

Cable	Pinout	Contacts covered
Cable 1	Side A pin 1 to Side B pin 3 Side A pin 2 to Side B pin 4	J1 and J4
Cable 2	Side A pin 1 to Side B pin 3 Side A pin 2 to Side B pin 1	J1 and J2
Cable 3	Side A pin 1 to Side B pin 3 Side A pin 2 to Side B pin 2	J1 and J3
Cable 4	Side A pin 1 to Side B pin 1 Side A pin 2 to Side B pin 2	J2 and J3
Cable 5	Side A pin 1 to Side B pin 1 Side A pin 2 to Side B pin 4	J2 and J4
Cable 6	Side A pin 1 to Side B pin 2 Side A pin 2 to Side B pin 4	J3 and J4

Plug types

Cables 1-4 use a 2-pin Side A plug and a 4-pin Side B plug. Cables 5-6 use 4-pin plugs on both ends; leave Side A pins 3 and 4 open.

For panel wiring and GPIO details, see hardware/raspberry-pi/README.md in the project repository.